

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Operating Systems [ECM_3123]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given the following set of processes, their arrival times, burst times, and priorities: (5)

A)

Process	Arrival Time	Burst Time	Priority
P1	0	5	-1
P2	1	3	-2
P3	2	8	3
P4	3	6	1
P5	4	5	2
P6	5	2	-2

- a) Draw the Gantt chart for preemptive priority scheduling (if two processes have the same priority, schedule the earliest arrival). (Lower number indicates Lower priority)
b) Draw the Gantt chart for LRTF (Least Remaining Time First)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Consider the following code where two processes, proc1 and proc2, concurrently update a shared integer variable counter initialized to 0: (5)

```
int counter = 0;
proc1 {
  int X = counter;
  counter = X - 1;
  X = counter;
  counter = X + 1;
}
```

```
proc2 {
  int Y = counter;
  counter = Y - 1;
}
```

1. What is the expected final value of counter if the operations of proc1 and proc2 are executed

sequentially, without interruption or overlapping?

2. List all possible final values of counter when proc1 and proc2 execute concurrently without any synchronization.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2) Given the following data for a system with 6 processes (P0 to P5) and 4 resource types (A, B, C, D): (5)

A)

Process	Allocation	Maximum
	A B C D	A B C D
P0	2 1 3 1	4 3 4 2
P1	1 0 1 2	2 2 2 3
P2	3 2 1 0	6 3 3 1
P3	1 1 2 2	3 3 4 3
P4	0 0 2 1	2 2 3 2
P5	1 2 0 1	3 3 2 2

Available resources:

A = 4

B = 2

C = 2

D = 3

a) Construct the Need matrix for the given data.

b) Using the Banker's Algorithm, determine whether the system is currently in a safe state. If so, provide safe sequence(s).

c) If process P2 requests 1 instance of resource type A and 1 instance of D, can this request be granted immediately? Justify using Banker's Algorithm.

Course Outcome: "CO4" / Cognitive Level: "Remember"

- B) A system contains 5 processes (P1, P2, P3, P4, P5) and 3 resource types (R1, R2, R3) with the following number of instances: (5)

- R1: 4 instances
- R2: 3 instances
- R3: 3 instances

At a particular moment, the system's resource allocation status is:

- P1 holds 2 instances of R1 and is requesting 1 instance of R2.
- P2 holds 1 instance of R2 and is requesting 2 instances of R3.
- P3 holds 2 instances of R3 and is requesting 1 instance of R1.
- P4 holds 1 instance of R2 and is not requesting any additional resources.
- P5 is requesting 1 instance of R1 and 1 instance of R3 but holds none currently.

a) Draw the resource allocation graph showing all allocations and requests.

b) Is the system currently in a deadlock state? Justify your answer based on the RAG.

c) If one instance of R3 is released by P3, explain how the graph and process states change, and identify which processes can proceed.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3) A computer system uses paging for memory management. (5)

A)

The system has a 36-bit logical address space.

The page size is 8 KB.

The physical memory is 512 MB.

- a) How many bits are there in the logical address used for the page number and for the page offset?
- b) How many page frames are there in physical memory? How many bits are required for the physical frame number?
- c) Given the following page table for a process:

Page Number	Frame Number
0	3
1	7
2	14
3	29

Translate the logical address 0x0001ACF4 to a physical address for this process.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Consider a computer system with a virtual address size of 36 bits and a page size of 8 KB. Each page table entry occupies 4 bytes, and each page table must fit exactly into one page frame. Find how many levels of page tables are necessary to accommodate this structure. Demonstrate how the virtual address and physical address are divided into page table indices and offsets in this multi-level paging system. Finally, calculate the size of the physical memory based on these parameters. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4) Given a system with 3-page frames and the following page reference string: (5)

4,1,2,4,3,2,1,5,4,2,1,6,3

- A)
- Simulate the First-In-First-Out (FIFO) page replacement algorithm and determine the number of page faults. Draw the state of the page frames after each page reference.
 - Simulate the Most Recently Used (MRU) page replacement algorithm on the same reference string and calculate the number of page faults.
 - Compare the number of page faults from FIFO and MRU for this reference string and explain which algorithm performs better and why.

Course Outcome: "CO5" / Cognitive Level: "Remember"

- B) Consider a system in which the time taken to access the Translation Lookaside Buffer (TLB) is 40 nanoseconds, and the main memory access time is 100 nanoseconds. The system has a TLB hit ratio of 0.85, meaning 85% of memory references successfully find the required page table entry in the TLB. Additionally, the page fault rate is 0.004, and each page fault requires 8 milliseconds to be serviced, including disk I/O. Using this information, calculate the Effective Memory Access Time (EMAT) for the system by accounting for TLB hits, TLB misses, and the overhead caused by page faults. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- 5) A magnetic disk has the following characteristics: (5)

- A)
- Number of tracks: 1000
 - Average seek time per track = 0.1 ms
 - Disk rotation speed = 7200 RPM (rotations per minute)
 - Data transfer rate = 100 MB/s
 - Size of data block to be transferred = 4 KB
 - Controller overhead time = 1 ms

Calculate: Total average disk access time

Course Outcome: "CO6" / Cognitive Level: "Analysis"

- B) A disk has 156 tracks. The disk arm is currently at track 44. The queue of requests in order is: (5)
89,129,11,123,51,9,58,120
Using the above, answer the following:
a) Calculate the total head movement (seek time) and show the order of servicing requests using:
- Shortest Seek Time First (SSTF)
 - Elevator Algorithm
 - C-SCAN Algorithm
 - LOOK Algorithm
 - C-LOOK Algorithm
 - SCAN Algorithm
- b) Which algorithm provides the best average seek time for this request set? Justify briefly

Course Outcome: "CO6" / Cognitive Level: "Remember"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	2	CO 2	5
2	3	CO 3	5
3	4	CO 4	10
4	5	CO 5	20
5	6	CO 6	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	3	15(30%)
2	Understand	0	0(0%)
3	Application	3	15(30%)
4	Analysis	4	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)